CLEARWATER VALLEY IRRIGATION DISTRICT

2024 ANNUAL FACILITY CONDITION ASSESSMENT

Cottonwood Creek Pumping Plant No. 2 (PP-2)
Clearwater Valley Regional Water Conveyance System

Facility:	Cottonwood Creek Pumping Plant No. 2 (PP-2)
Location:	NE1/4 Sec. 22, T4N, R2W, B.M., Elmore County, Idaho
Year Constructed:	2019 (Phase 1 Construction)
Original Design:	Western Geotechnical Consultants, Report 2018-GEO-0093
Inspection Date:	August 22, 2024
Inspector:	T. Frederickson, P.E., CVID Facilities Engineer
Accompanied by:	M. Salazar, CVID Operations Superintendent
Weather:	Clear, 78°F, dry conditions
Report Date:	September 5, 2024

1.0 EXECUTIVE SUMMARY

The Cottonwood Creek Pumping Plant No. 2 (PP-2) was inspected on August 22, 2024, as part of the District's annual facility condition assessment program. The facility has been in continuous seasonal operation since commissioning in June 2019 (five full irrigation seasons). Overall, the facility is in **good condition** and performing as designed. No structural deficiencies, foundation distress, or geotechnical performance issues were identified.

Minor maintenance items were noted, including hairline cosmetic cracking in the pump bay interior walls and minor erosion at the site drainage outfall. These items are classified as routine maintenance and do not affect the structural integrity or operational capacity of the facility. No immediate corrective action is required; the items are recommended for inclusion in the District's 2025 maintenance work plan.

2.0 FACILITY DESCRIPTION

PP-2 is a reinforced concrete pump station housing three 500-horsepower vertical turbine pumps with a combined design capacity of 15,000 gallons per minute. The facility was constructed in 2019 based on the geotechnical investigation and recommendations provided by Western Geotechnical Consultants (WGC Report No. 2018-GEO-0093, October 2018). The principal structural and geotechnical elements include:

- **Pump Bay:** Reinforced concrete below-grade structure, approximately 45 ft x 25 ft in plan, extending 12 ft below original grade. Founded on drilled shafts (6 shafts, 20-inch diameter) socketed 5 to 6 feet into the underlying Challis Volcanic bedrock at approximately 21 to 23 ft below grade.

- **Motor Control Center / Electrical Room:** At-grade reinforced concrete slab and masonry block structure, founded on spread footings bearing on the alluvial clay stratum (Layer 2) with 24-inch compacted structural fill pad.
- **Discharge Manifold:** 36-inch reinforced concrete pipe manifold connecting to the transmission main, supported on spread footings.
- **Access Road and Retaining Wall:** Reinforced concrete retaining wall (6 ft exposed height) along the south perimeter, with compacted gravel access road.

3.0 INSPECTION FINDINGS

3.1 Structural and Foundation Performance

The pump bay structure shows no evidence of differential settlement, tilting, or foundation distress. Floor slab elevations were checked at four locations using a digital level and compared to the as-built survey (July 2019). Maximum measured deviation from the original survey is 0.08 inches (less than 1/8 inch), which is within the range of measurement precision and indicates negligible settlement over the five-year service period. This is consistent with the drilled shaft foundation system bearing on competent bedrock.

The motor control center building slab was similarly checked at two corner locations. Measured deviations from as-built were 0.15 and 0.22 inches, indicating minor settlement well within the 0.75 to 1.25 inch range predicted in the original geotechnical report. No distress to the slab, walls, or door frames was observed. Settlement appears to have stabilized, consistent with the prediction that primary consolidation would be substantially complete within 6 to 12 months of construction.

3.2 Concrete Condition

The pump bay interior walls exhibit several hairline cracks (less than 0.01 inch width) on the north and east walls between elevations 5,472 and 5,474 ft. The cracks are predominantly horizontal and appear to follow construction joints or shrinkage patterns. No staining, efflorescence, or evidence of water infiltration was observed at any of the crack locations. The cracks are classified as **cosmetic** and do not indicate structural distress or reinforcement corrosion.

The retaining wall concrete is in good condition with no cracking, spalling, or deterioration observed. The wall shows no evidence of outward rotation, lateral displacement, or backfill settlement behind the wall. Weep holes at the base of the wall were clear and functioning. No evidence of sulfate attack or corrosion-related distress was observed in any below-grade concrete, indicating that the Type II cement and corrosion protection measures specified in the geotechnical report are performing as intended.

3.3 Swell Performance

No evidence of swell-related distress was observed at any structure. The moisture barriers installed around the spread footings appear to be intact based on visual observation of exposed edges at grade. The 24-inch structural fill pad beneath the spread footings is performing its intended function of isolating the foundation from the underlying expansive clay. No floor slab heaving, door binding, or wall cracking patterns consistent with swell distress were noted.

The void space beneath the grade beams connecting the drilled shafts in the pump bay was verified at two accessible locations by probing with a rod. The void remains open (approximately 3 to 4 inches of clearance), confirming that the clay subgrade has not swelled sufficiently to close the gap. This is an important positive finding and should continue to be monitored annually.

3.4 Groundwater and Drainage

The pump bay sump was dry at the time of inspection. Operations staff reported that minor seepage into the sump occurs during peak irrigation season (May-June) when groundwater levels are highest, requiring intermittent sump pump operation (estimated 1 to 2 gallons per minute). This is consistent with the seasonal groundwater fluctuation predicted in the geotechnical report and does not indicate a drainage deficiency.

The retaining wall drainage system (perforated pipe and granular backfill) is functioning properly with no evidence of hydrostatic pressure buildup or wall seepage. The site surface drainage is generally adequate, with positive grades directing runoff away from the structures. However, minor erosion was noted at the drainage outfall on the north side of the site where concentrated flow has created a small gully (approximately 12 inches deep by 18 inches wide by 8 feet long) in the native topsoil. This should be repaired with riprap or an energy dissipater to prevent progressive erosion toward the pump bay.

3.5 Site Conditions and Grading

The access road surface is in fair condition with minor rutting in the unpaved gravel section (approximately 1 to 2 inch ruts). The paved apron at the pump bay entrance is in good condition with no cracking or settlement. Vegetation management around the facility is adequate. No erosion, slope instability, or evidence of burrowing animal activity was observed on the excavation backfill slopes.

The groundwater monitoring standpipes installed during the original 2018 investigation were checked. BH-PP2-01 and BH-PP2-04 standpipes are still accessible and functional. Groundwater was measured at 15.8 ft bgs in BH-PP2-01 and 16.1 ft bgs in BH-PP2-04 on August 22, 2024 (late irrigation season). These readings are approximately 2 feet lower than the September 2018 measurements (13.5 and 14.0 ft, respectively), consistent with normal seasonal and annual variation. BH-PP2-02 and BH-PP2-03 standpipes were damaged during construction and are no longer serviceable.

4.0 CONDITION RATING SUMMARY

Table 1: Component Condition Ratings

Component	Condition Rating	Change from 2023	Action Required	Notes
Pump bay structure	**Good**	No change	None	No settlement, no cracking
Drilled shaft foundations	**Good**	No change	None	Negligible settlement (<0.1 in)
Spread footings (MCC)	**Good**	No change	None	Settlement 0.15-0.22 in, stable
Pump bay concrete	**Good**	No change	Monitor	Hairline cracks, cosmetic only
Retaining wall	**Good**	No change	None	No displacement, weep holes clear
Swell mitigation	**Good**	No change	Monitor	Void space verified at 3-4 in
Corrosion protection	**Good**	No change	None	No sulfate attack or deterioration
Groundwater/drainage	**Good**	No change	Repair outfall	Minor erosion at N outfall
Access road	**Fair**	Slight decline	Grade and add gravel	Minor rutting in gravel section
Site grading/vegetation	**Good**	No change	None	Positive drainage, adequate cover
GW monitoring wells	**Fair**	No change	Replace PP2-02, -03	2 of 4 damaged in construction

Rating scale: Good = no deficiencies, performing as designed; Fair = minor deficiencies, routine maintenance needed; Poor = significant deficiencies, corrective action required.

5.0 RECOMMENDATIONS

5.1 Routine Maintenance (2025 Work Plan)

- **Drainage outfall repair:** Install riprap energy dissipater (approximately 4 ft x 6 ft x 12 inches deep, Class 2 riprap) at the north drainage outfall to arrest gully erosion. Estimated cost: $800 - $1,200.

- **Access road maintenance:** Grade and add 2 inches of 3/4-inch crushed gravel to the unpaved section (approximately 200 linear feet). Estimated cost: $1,500 - $2,500.

- **Monitoring well replacement:** Consider replacing the damaged BH-PP2-02 and BH-PP2-03 standpipes during the 2025 or 2026 supplemental investigation for the proposed holding tank (TK-2A). This would restore the full groundwater monitoring network at minimal incremental cost if combined with a drilling mobilization.

5.2 Continued Monitoring

- Continue annual floor elevation surveys at the pump bay (4 points) and MCC building (2 points) to confirm settlement has stabilized.

- Continue annual grade beam void space verification at the pump bay. If the void space decreases below 2 inches at any location, notify the District Engineer and consult with WGC regarding supplemental swell mitigation measures.

- Continue semi-annual groundwater level measurements in BH-PP2-01 and BH-PP2-04 (spring and late summer readings). Maintain the monitoring record to support future design work, including the proposed TK-2A holding tank project.

- Monitor the hairline cracks in the pump bay walls. If any crack exceeds 0.02 inches in width, exhibits staining or efflorescence, or shows evidence of active water infiltration, consult with a structural engineer.

5.3 Future Capital Projects

The District is currently evaluating the addition of a surge/holding tank (TK-2A) adjacent to the existing PP-2 facility as part of Phase 2 system improvements. Based on the consistent geotechnical conditions documented across the PP-2 site during the original 2018 investigation and confirmed during 2019 construction, a supplemental geotechnical investigation of limited scope (one to two borings) should be adequate to characterize conditions at the proposed tank location, provided the tank site is within approximately 500 feet of the existing borings and on the same alluvial terrace surface.

The five years of satisfactory foundation performance at PP-2, combined with the consistency of subsurface conditions observed across four borings and during construction, provides a strong basis for extrapolating the existing geotechnical model to the nearby tank site. The original WGC report (2018-GEO-0093) and the 2019 construction inspection memorandum should be provided to the geotechnical consultant performing the supplemental investigation for reference.

6.0 OVERALL ASSESSMENT

The Cottonwood Creek Pumping Plant No. 2 is in good overall condition after five years of seasonal operation. The geotechnical design, including the drilled shaft foundations, spread footings with swell mitigation, and corrosion protection measures, is performing as intended. Foundation settlement is within predicted limits and appears to have stabilized. No geotechnical performance issues were identified that would affect the continued safe and reliable operation of the facility.

The facility condition rating for 2024 is **GOOD** (9 of 11 components rated Good, 2 rated Fair). The Fair-rated items (access road and monitoring wells) are routine maintenance concerns and do not affect the structural or geotechnical performance of the facility.

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